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Review

Beyond Composting Basics: A Sustainable Development Goals—Oriented Strategic Guidance to IoT Integration for Composting in Modern Urban Ecosystems

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Abstract: IoT-based composting provides clear advantages over conventional urban composting in areas such as enhanced monitoring, efficiency, resource utilization, and management. Bibliometric analysis of 121 publications on IoT-based urban composting identified critical research gaps and emphasizes the necessity for a strategic framework for full implementation and execution of sustainable development goals-oriented IoT-based composting in modern cities across. Under the key theme of IoT-based urbanized composting automation, 16.5% of publications focus on urbanized composting automation but overlook the system's scalability. The lowest mean citations of 72.7 (22.3% of publications) in intelligent composting process optimization show the lack of broader applications. A total of 28.9% of total publications focus on urban composting sustainability assessment but lack IoT integration in their scope. The composting process, pollution, environmental impact, cost, and life cycle analysis of modern city composting share 19% and 13.3%, respectively. However, both key themes lack real-time monitoring, operation, and economic feasibility for scalable models. The article highlights a fragmented landscape providing sustainable development goals-oriented strategic guidance for the full implementation and execution of IoT-based composting facilities in modern city ecosystems. The article comprehensively explains the budgetary constraints, scalability, data management, technological compatibility, privacy, security, and regulatory compliance essential for sustainable operation.

Keywords: internet of things; composting; sustainable development goals; life cycle assessment; inclusivity; community engagement; waste conversion hotspots

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1. Introduction

The increase in the demand for food and sanitation has led to the accumulation of compostable waste, such as yard waste, disposable paper products, and food waste in urban areas. As an eco-friendly solution, such compostable waste can be transformed into compost. The transformation of such compostable waste into compost can rapidly recycle the nutrients and the chemical elements of such waste to be used in agriculture and soil amendment in a shorter time [1–4]. Despite the rapid accumulation of compostable waste, modern cities must attain city-wide compostable waste sustainability through a circular economy, such as resource conservation, environmental impact reduction, adherence to responsible production and consumption, business sustainability, and inclusive industrial development.

Due to the large volume of compostable waste generated in modern cities and the lack of labor to handle the compostable waste, conventional urban composting facilities lag. Therefore, to handle the incomprehensible amount of compostable waste, automation of the composting process via successful integration of the Internet of Things (IoT) and related technologies with the conventional composting process is a successful strategy [5–9]. Even though sporadic integration of IoT and other related technologies, such as mobile apps, are adopted for individual composting processes, there is a lack of comprehensive strategic guidance for the digitization of composting in modern cities [1,10–13].

Many studies emphasize the benefits of IoT-based composting facilities in a modern city ecosystem. However, proper guidance for an actionable plan and framework for the successful implementation of IoT-based composting in smart cities is hardly found. The rapid increase in the IoT integration into waste management operations worldwide, considerable inefficiency in conventional urban composting plants based on outdated sensors, and lack of novel state-of-the-art IoT components in modern urban composting prompted the development of the review to develop a strategic framework for fully functional IoT-based composting facilities in urban settings.

Ref. [14] defines a modern city ecosystem as an urban area leveraging advanced technologies, particularly IoT, to improve the quality of life for its residents, promote sustainable development, and improve urban services. Modern city ecosystems utilize interconnected devices and analytics to optimize energy management, waste management, transportation, and public safety functions. Therefore, this paper focuses on an integral part of the modern city ecosystem, which is urban waste composting. As described by [14], this is one of the vital facets of modern city ecosystems for an efficient, livable, and sustainable environment via IoT integration and active citizen engagement. Governments worldwide are setting mandates such as the Organic Waste Diversion Law of California to encourage the adoption of smart bins and IoT-based real-time waste management monitoring [15,16]. Modern cities with zero waste goals already use IoT-based composting to track progress and optimize urban waste collection [17,18]. For example, the U.S. Environmental Protection Agency (EPA) also drives IoT adoption in composting to regulate greenhouse gas emissions in U.S. cities [19]. Even though regulatory and compliance pressure is on the rise in utilizing IoT-based composting solutions, a lack of proper guidance for an actionable plan and framework for the successful implementation of IoT-based composting in smart cities is hardly found. The rapid increase in the IoT integration into waste management operations worldwide, considerable inefficiency in conventional urban composting plants based on outdated sensors, and lack of novel state-of-the-art IoT components in modern urban composting prompted the development of the review to develop a strategic framework for fully functional IoT-based composting facilities in urban settings.

Therefore, a thorough systematic review of more than 100 publications (journal articles, conference proceedings, book chapters, and technical reports) was done to define scope and conduct bibliometric analysis to identify the research gaps and trends. Then the article includes essential counterparts of a scalable and seamless IoT-based composting process in a city ecosystem, followed by factors to consider for adopting an IoT-based compost automation system in modern cities. Finally, the article discusses step-wise integration of an IoT-based composting setup to a modern city and its ecosystem and guidance for achieving sustainability on SDGs during deployment of the IoT-based composting facility.

The objective of this review is manifold. First and foremost, the authors investigate the essential components for an integrative, upgradable, and seamless IoT-based composting facility, highlighting the edge and unique benefits of such integration, such as improved city productivity and efficiency. A crucial part of the review also evaluates the factors to be considered in effectively implementing IoT and related technologies into composting in a sustainable framework. These factors include problems with scalability, data privacy, security, architecture, and governance. Overall, the review provides strategic guidance for implementing IoT-based composting facilities in modern city ecosystems by reducing real-world obstructions such as improper alignment with SDGs, community disagreements, and financial viability. Readership can get a clear understanding of the structure of the article by referring to Figure 1 below.

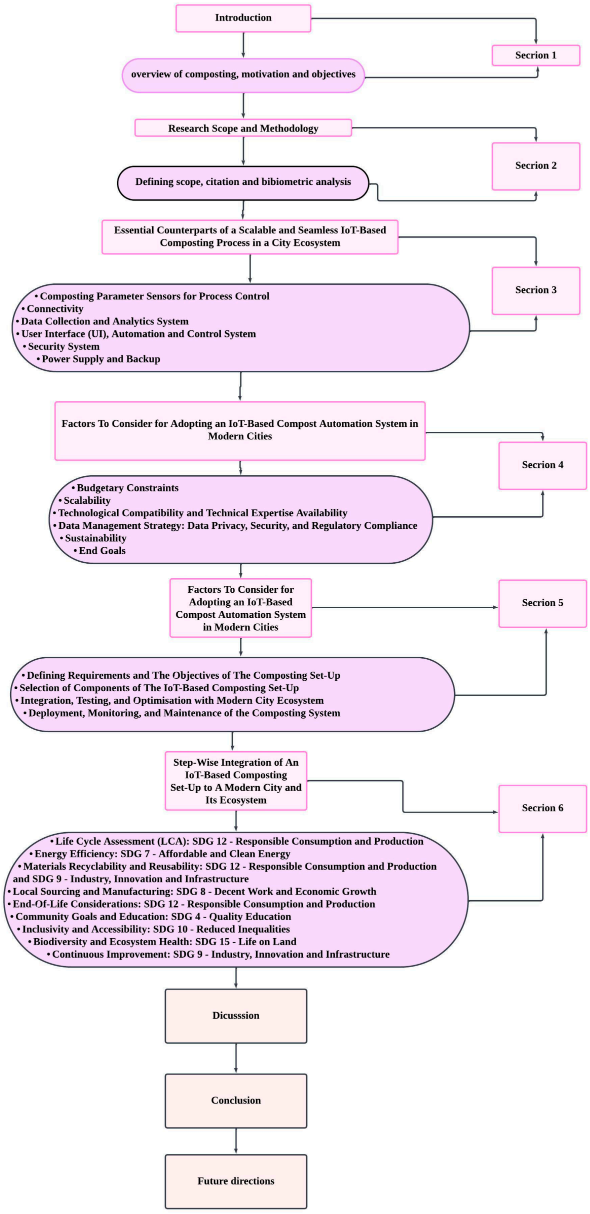

Figure 1. Overview of the article structure.

2. Research Scope, Methodology, Bibliometric, and Citation Analysis

2.1. Scope of the Study

The review aims to critically analyze the development and IoT integration into modern city composting systems in a sustainable framework. By analyzing 121 publications across five key themes: (1) IoT-based urbanized composting automation; (2) intelligent composting process optimization; (3) urban composting sustainability assessment; (4) composting process pollution and environmental impact; (5) cost and life cycle analysis of modern city composting.

2.2. Research Methodology

The primary goal of the systematic review is to give a comprehensive understanding of the current state of the art on a research topic or a section. The systematic review was carried out as described by (author). Two primary databases were considered for the systematic review: (1) SCOPUS (Elsevier Publishers) and (2) Web of Science (Institute for Scientific Information/Clarivate Analytics). As the search covers major scientific publications, we have confidence that all the major research approaches for creating a strategic, sustainable IoT-based composting facility in modern city ecosystems. The research criteria followed a TITLE-ABS-KEY schema using keyword terms' internet of things', 'smart waste management', 'intelligent waste sorting', 'composting automation', 'real-time monitoring', 'IoT-based composting', 'urban waste management', 'waste solutions', 'composting systems', 'organic waste recycling', 'machine learning', 'augmented reality', 'computer vision', 'artificial intelligence', 'sensor-based systems', 'data-driven decision-making', 'circular economy', 'sustainable development goals', 'pollution discharge monitoring', 'cost analysis', 'life cycle assessment', 'modern city'. The initial articles were filtered using pre-determined criteria for eligibility and relevance. The exclusion criteria for the articles include studies on non-composting waste management (e.g., electronic waste, hospitals, and industrial), outdated or non-specific research, general IoT applications non-related to composting, and non-peer-reviewed articles with non-urban focus unless applicable to urban settings. One hundred twenty-one were found to be directly related to the comprehensive IoT-based technology integration into sustainable modern city waste management. Next, the selected works were organized into categories. According to the scope of the study, 121 publications were analyzed across five key themes: (1) IoT-based urbanized composting automation; (2) intelligent composting process optimization; (3) urban composting sustainability assessment; (4) composting process pollution and environmental impact; (5) cost and life cycle analysis of modern city composting.

2.3. Bibliometric and Citation Analysis

A bibliometric analysis of 121 references shows critical insights into the existing research landscape for IoT-based composting in modern city ecosystems. It emphasizes the importance of a cohesive and strategic approach. Journal publications dominate 80.99% of the total; 14.05% are conference proceedings, while only 4.96% are book chapters and technical reports (Figure 2). This suggests that despite a solid academic foundation, more applied and practical guidance for scalable solutions is needed (Figures 3 and 4).

16.5% (20 publications) are focused on IoT-based urbanized composting automation, reflecting increased attention to automation and real-time urban composting process monitoring with limited attention to the scalability of the systems (e.g., [20,21]). With an average of 194.8 citations per publication, IoT-based urbanized composting automation shows high variance, signaling fragmented research in urban-level applications. Intelligent composting process optimization comprises 22.3% (35 publications), with the lowest citation impact of 72.7 of mean citations. Therefore, despite the development of localized intelligent composting process optimization systems, there is a need for more cohesive frameworks optimized for city-wide composting operations (e.g., [22,23]). Studies like [20,24] are included in the largest category, sustainability and urban waste management, which accounts for 28.9% of all publications (35 total), with an average of 138.5 citations per article. Despite widespread interest in sustainability, only a few of these studies integrate IoT to achieve sustainability in urban ecosystems, exposing limitations in adopting IoT. As seen in [25,26], pollution monitoring and environmental impact, which accounts for 19.0% (23 articles), has a strong emphasis on pollution control. It has an average of 163.5 citations but little IoT integration for real-time environmental monitoring. Lastly, with an average of 211.8 citations, cost and life cycle assessment accounts for 13.3% of all papers (16 publications), including works such as [27,28]. Although economic evaluations are of great importance, no reliable cost models in this area are necessary for expanding IoT composting systems in urban environments. Figure 3 shows the span of the publications across key themes. Table 1 shows the citation analysis across key themes of the systematic literature review.

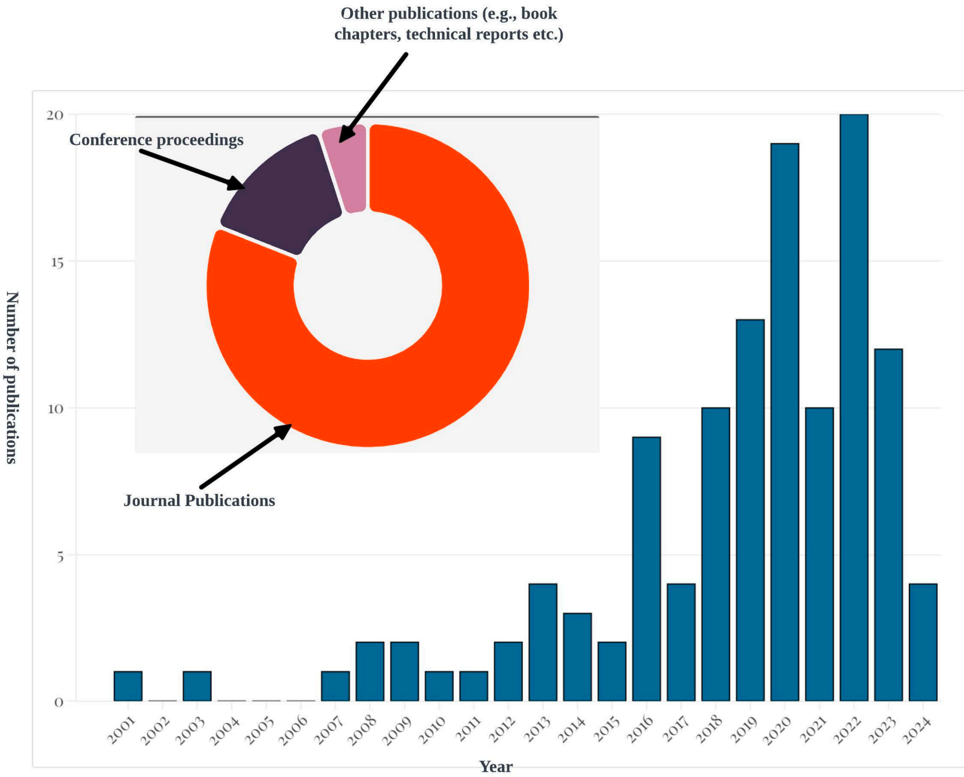

Figure 2. Publication type categorization of the research performed in the field and the number of publications per year from 2001 to 2024.

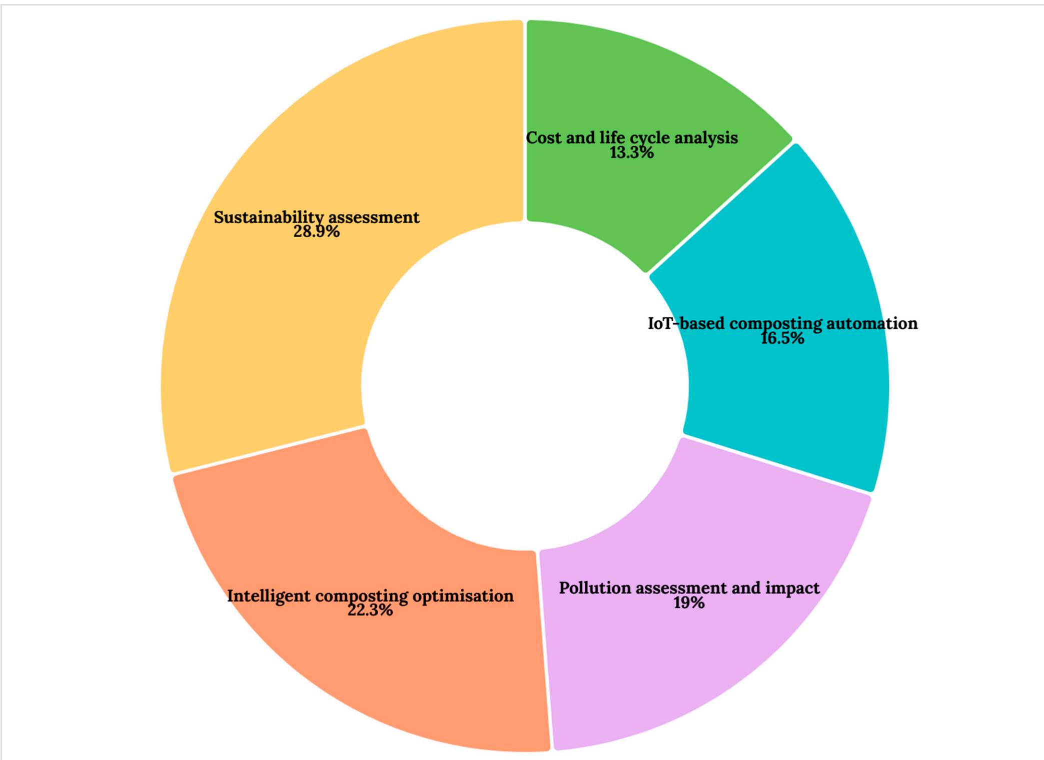

Figure 3. Spanning of the publications under key themes of the review as a percentage.

Table 1. An overview of citation metrics analysis of the references used in the systematic review.

Key Theme	Total Citations	Mean Citations	Standard Deviation	Mean Citations per Year	Standard Deviation of Citations per Year
IoT-based urbanized composting automation	3896	194.80	±631.95	24.34	±50.26
Intelligent composting process optimization	4849	138.54	±343.34	28.49	±64.62
Urban composting sustainability assessment	1964	72.74	±188.74	10.53	±14.59
Composting process pollution and environmental impact	3762	163.56	±306.93	28.85	±40.77
Cost and life cycle analysis	3177	211.80	±306.72	23.22	±35.58

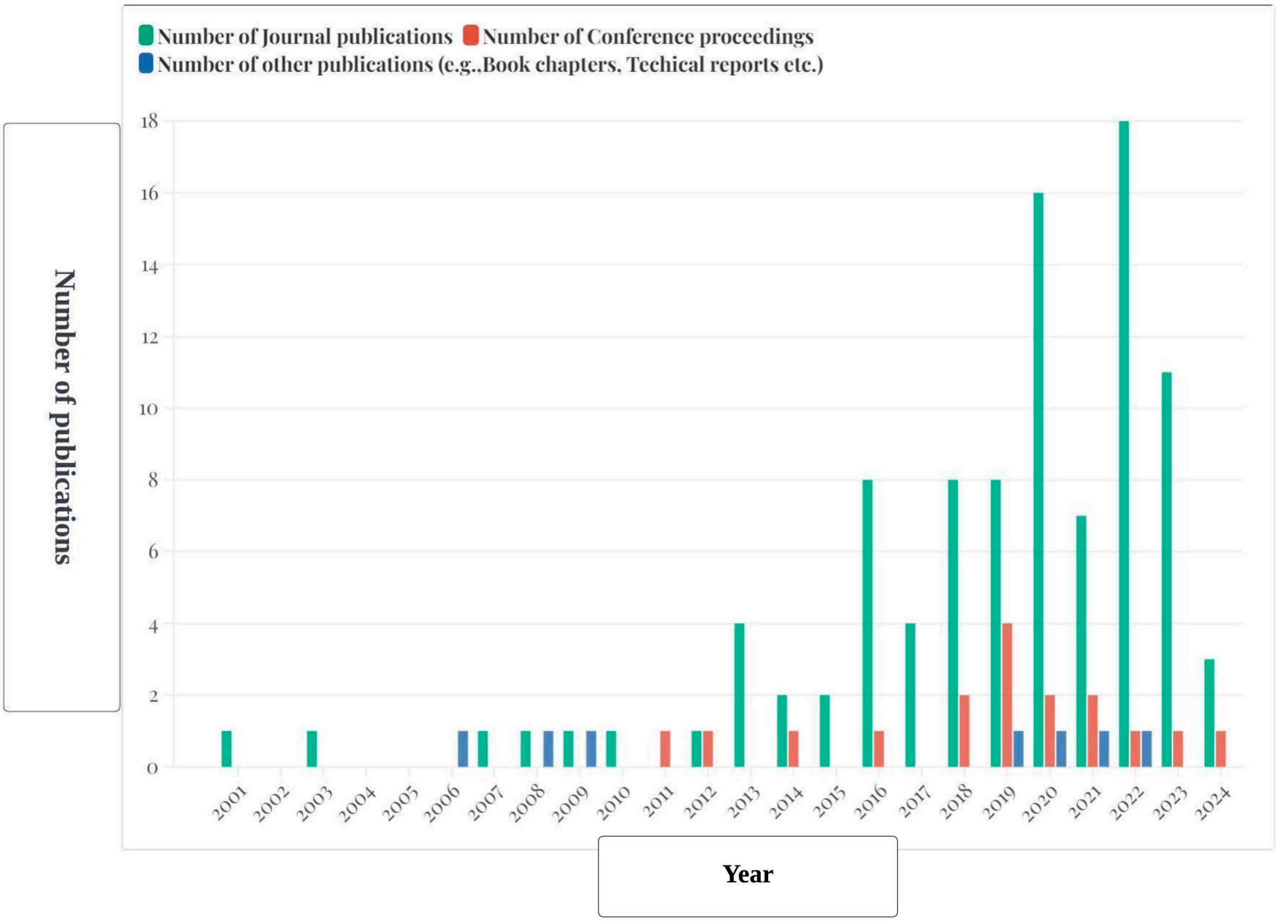

Figure 4. The number of journal publications, conference proceedings, and other publications (e.g., book chapters, technical reports, etc.) over the time span of 2001 to 2024.

Gaps in converting theoretical developments into practical frameworks, as evidenced by the dispersed approach across themes, significant citation variation, and minimal representation of applied research (few conference proceedings and technical reports). In order to promote complete IoT-based composting solutions in smart cities, this research emphasizes the necessity for standardized, scalable IoT frameworks that incorporate sustainability, artificial intelligence, pollution monitoring, and economic models.

3. Essential Counterparts of a Scalable and Seamless IoT-Based Composting Process in a City Ecosystem

3.1. Composting Parameter Sensors for Process Control

IoT-based composting process parameters sensors can detect and measure vital parameters such as moisture level, pH, temperature, and oxygen concentration. The collected data are critical to understanding the ongoing decomposition conditions and to optimizing the conditions for an accelerated decomposition. For example, smart composting tumblers equipped with composting process parameter sensors can obtain real-time data on organic matter decomposition [29]. This data feed can be used to capture real-time data of different composting stages and allow operators to perform precise adjustments to optimize the thermophilic phase and accelerate the transition from the mesophilic phase to the thermophilic phase [20]. This data feed from the composting process parameter sensors can be used as training data for computer vision, AR (Augmented Reality), AI (Artificial Intelligence), and neural networks [30]. Therefore, IoT-based composting process parameter sensors can serve a dual purpose in composing. One is parameter monitoring and detection of various counterparts that emanated during decomposition and serve as raw data platforms for the development of more advanced algorithms for composting, such as computer vision-enabled waste segregation.

Machine learning (ML), AI, and computer vision significantly enhance compost production in IoT-based composting facilities by enhancing real-time composting monitoring, intelligent waste sorting, and process optimization. Intelligent waste segregation can be achieved by computer vision, which uses deep learning models to identify and segregate compostable urban waste from non-compostable materials accurately. This helps reduce contamination and enhances final product quality, as demonstrated by [24]. Real-time composting conditions can be monitored using ML algorithms that use data feeds from sensors tracking oxygen levels, moisture, and temperature and allow dynamic adjustments to optimize microbial decomposition, accelerate composting, and assure high-quality end products [22]. Furthermore, artificial neural networks (ANNs) can be effectively used for predictive modelling of compost quality, where historical data from composting processes can forecast nutrient content and maturity. A study by [31] demonstrated that ANN was employed to predict NH_3 emissions during composting with an R^2 of 0.97 using pH, electrical conductivity, temperature, carbon/nitrogen ratio, and ammoniacal nitrogen levels. This allows optimized composting with minimal harmful emissions. A study by [32] Utilized AR glasses to detect waste in real-time without user interaction to capture images. This approach allowed users to classify urban waste into compostable and non-compostable via 3D object detection with minimal reaction time.

IoT-based sensors can be effectively used for real-time constant monitoring and control of critical composting parameters such as pH of 6.0 to 8.0 (electrode pH sensors), moisture content of 50% to 60% (capacitive or resistive moisture sensors), the temperature of 55 °C to 65 °C during thermophilic phase (thermocouples or fiber optic sensors (FOS)), aeration levels of 0.4 to 0.6 L/(min·kg organic matter) during the initial stages, decreasing to 0.2 to 0.4 L/(min·kg organic matter) during maturation (electrochemical/optical oxygen sensors), conductivity, and metal oxide semiconductor sensors (MOSs) for leachate monitoring and pollution control with minimal human supervision [33–36]. Furthermore, deep learning approaches can automate and optimize composting parameters. Already existing composting optimization and automation algorithms by [37] can be used to identify all the types of physio-chemical constraints recurring in the feedstock and considered simultaneously during the composting process. A great deal of control over pollution control can be achieved using MOS via gaseous emissions detection of NH_3, CO_2, and volatile organic compounds (VOCs). When emissions exceed a predetermined threshold, the system can automatically adjust the aeration rate and activate odor control measures such as biofilters [38]. During composting, the quality of leachate is monitored for detecting harmful substances such as metals (Pb, Cu, Zn, Fe, Mn, Cr, Cd) and nutrients (NO_3^-, NO_2^-, NH_4^+) that might leach from the compost pile. Conductivity sensors can monitor leachate quality and trigger a discharge control mechanism, pretreating or recirculating the leachate via coagulation and membrane separation to prevent pollution [39]. The urban composting optimization algorithm proposed by [37] can be used to define the plant's maximum and minimum capacity constraints. The minimized formulation of the expression of such a mathematical expression is as follows:

$$\text{Minimize}: \sum_{i=1}^{n} c_i\, x_i + \sum_{i=1}^{n} \sum_{j=1}^{k} m_j y_{ji} + \sum_{i=1}^{n} r_i z_i \tag{1}$$

where: i = current mixture; j = current raw material;(unknown factors): x = amount of each mixture; y = amount of each raw material for each mixture; z = quantity of each other resource required for each mixture; (known parameters): n = number of mixtures that it is assumed undergo the composting process; k = number of raw materials used in the process; c = overall unit cost (other than the supply of raw materials) of each mixture; m = unit cost of each raw material; r = unit cost of each other resource.

The mathematical expression can be further minimized to different constraints to evaluate the maximum plant capacity constraints, and maximum or minimum raw material availability.

For maximum plant capacity constraints.

$$\text{Subject to}: \sum_{i=1}^{n} \sum_{j=1}^{n} y_{ji} \leq capacity \tag{2}$$

For maximum or minimum raw material availability:

$$\text{Subject to}: \sum_{i=1}^{n} y_{ji} \geq mininum\ capacity\ of\ y_{ji} \tag{3}$$

Or

$$\text{Subject to}: \sum_{i=1}^{n} y_{ji} \leq maximum\ capacity\ of\ y_{ji} \tag{4}$$

3.2. Connectivity

An IoT-based composting system demands a robust and dependable network architecture to facilitate a seamless transmission of composting data from different sensors to a centralized monitoring system [40]. These mandates utilize reliable connectivity methods such as Wi-Fi networks, cellular networks, or any other mode of IoT connectivity. The effective transmission of the real-time composting sensor data to a central monitoring network depends on uninterrupted connectivity [41]. Depending on the requirements in composting facilities and environmental conditions in which the IoT-based composting system is deployed, the type of connectivity needed will differ [42]. For example, Wi-Fi networks provide a stable connection to a small-scale composting facility but might need additional access points for large-scale facilities. On the other hand, cellular networks might incur recurring subscription costs for connectivity services [43]. Therefore, operators and planners of composting facilities can use connectivity technologies such as Long-Range Wide Area Network (LoRaWAN) to gain long-range communications with relatively low power consumption [44]. The selection of suitable connectivity assures the facility's timely and accurate transmission and communication.

3.3. Data Collection and Analytics System

The data collected by the sensors in the composting facility can be accumulated after they are communicated via connectivity networks. The initial collection of the raw data are done by the sensors [45]. This raw data should then be transmitted via the connectivity network to the data collection and analytics system for further analysis. The primary duty of the data collection and analytics system must be to compile, sort out, and analyze the raw data feed. Via advanced analytics systems such as AI and neural networks, actionable insights regarding decomposition can be given. These insights can then be used to recommend optimal conditions that promote compost maturity while pinpointing the issues and inefficiencies within the composting facility [46].

3.4. User Interface (UI), Automation, and Control System

The UI in an IoT-based composting system can be generally implemented as a mobile or web application. This allows the operators to access and interact with the processed data in real-time. Via the UI, operators can easily visualize different aspects of composting, such as the composting bin fill levels, real-time updates of feedstock collection schedules, decomposition rates, and compost sales figures [47]. Moreover, the UI allows the operator to oversee the composting process and provides users with alerts regarding anomalies that might occur during waste disposal or composting. This allows the expert technicians to immediately get their hands on any issue and direct the workers to correct it without being physically present [48]. Automation and control systems are an integral part of an IoT-based composting system, and they are crucial in making real-time decisions by leveraging the analytics of composting obtained from different sources [49]. Automation and control systems could control all the physical and controllable aspects of composting, such as pH, temperature, moisture, oxygen, etc. For example, the automation and control system can regulate the aeration to control the oxygen level in the composting feedstock. The automation and control system can adjust the aeration to optimize the oxygen levels best suited for decomposition by monitoring the sensor data fed by the oxygen or gas sensors in the system.

3.5. Security System

To ensure the protection of the data generated by the composting system due to the growing concern for privacy and cybersecurity and to safeguard industry secrets, it is essential to have a security system for the IoT-based composting system. Security systems can perform robust measures such as data encryption, access control, and security audits to ensure safety [50,51]. For example, the security system can convert the data generated during decomposition, research, and development into an unreadable format of any cryptography and block malicious entities from accessing the protected data. Another example is that security systems can implement authentication and authorization measures to ensure only authorized personnel access the data and interact with different counterparts, such as sensors, connectivity, and communication systems, to prevent tampering or misuse.

3.6. Power Supply and Backup

Above all, a reliable and uninterrupted power supply is essential to keep up the continuous operation of the IoT-based composting system [52]. This is especially important in data gathering, transmission, and processing. A robust power backup is needed when the power supply is imperative to avoid any disruptions due to power failures [53,54]. A reliable power supply ensures continuous operations in the composting facility, and failure to do so will result in hardware damage, system malfunctions, and data corruption. For example, different power backup methods, such as batteries or generated automatic transfer switches, can ensure the transition between primary and secondary power supplies in the composting plant. These components create an IoT-based composting system with seamless, efficient, and user-friendly features that enhance the overall composting process. After selecting the essential counterparts for a scalable and seamlessly operating IoT-based composting facility, the existing urban ecosystem's suitability for a smooth integration of the composting facility should be evaluated. The authors propose evaluating the following factors to lay out an action plan for implementing the composting facility. Figure 5 overviews the essential components of a scalable and upgradable IoT-based composting facility suitable for modern cities.

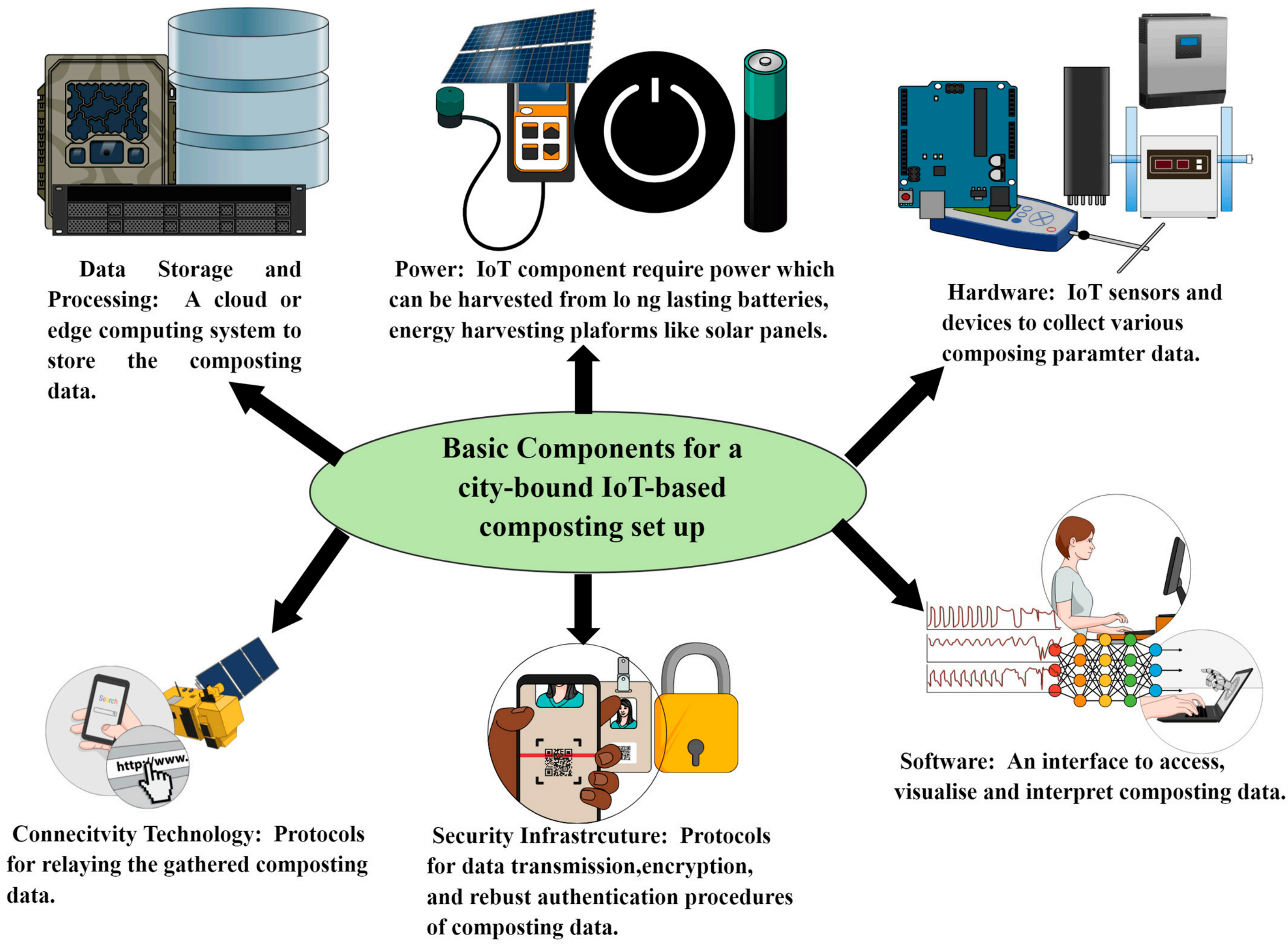

Figure 5. An overview of the essential components of a scalable and upgradable IoT-based composting facility suitable for modern smart cities.

4. Factors to Consider for Adopting an IoT-Based Compost Automation System in Modern Cities

Assessing the suitability of IoT integration into city-bound composting facilities depends on numerous factors, including environmental, social, and economic aspects. The authors propose the following factors for assessing the suitability of implementing IoT-based composting facilities in a selected city.

4.1. Budgetary Constraints

The budget required for IoT integration in terms of the installation and maintenance of these devices and their infrastructure should be the primary concern. Any composting facility intending to integrate IoT and related technologies into its ecosystem must intricately plan and ensure that sufficient financial resources are available for the initial integration and recurring expenses. The budget should include allocations for purchasing IoT devices such as sensors, modems, signal transmitters, and cameras, hiring skilled labor to install and maintain the IoT system, and investments for auxiliary but essential services such as data security. Moreover, operators should have enough allocations to train the existing staff on handling IoT concepts in composting and later managing it [55]. Table 2 directly compares the total costs of urban waste to compost using IoT-based composting facilities, conventional composting, and unsanitary and sanitary landfilling at the existing interest rates. Table 3 directly compares the total costs of conventional windrow composting, IoT-based automated facilities, sanitary landfilling, and unsanitary landfilling, with a 4% interest rate. The values were converted into present values of 2024 from 2019 using;

$$F = P(1+i)^n \tag{5}$$

where F and P stand for future and present values, respectively, i denotes discount rate (3%), and n denotes number of years [27].

Prior to installation of the facility, a life cycle cost (LLC) analysis needs to be done under the following assumptions [27,56].

$$Total\ cost = \left(CC \times \frac{i}{(1+i)^n}\right) + OC - OB + EC - EB \tag{6}$$

CC: Capital costs, including initial investment in land purchase, IoT sensors and components, power supply and backup, compost storage materials, and connectivity devices; i = interest rate; n = number of years; OC = Operational costs, including man hours, transportation, lubricants, maintenance, and utilities cost; OB = Operational benefits such as compost value; EC = Environmental costs: Environmental regulations costs (air pollution and water pollution); B = Environmental benefits, including avoided waste benefits, including carbon capture, CO_2 and CH_4 reduction. According to [57,58], the initial cost of the installation of IoT-based composting facilities is not proportional to the size of the composting facility due to their economic scale. According to [57], the cost of composting facility installation can be expressed as C, where the construction cost is C and facility treatment capacity (considered as t/day) is V.

$$C/V = \left[(C/C_0)(V/V_0)\right]^{-\propto} \tag{7}$$

Under the assumption of C = C_0 and V = V_0, the Equation (7) can be expressed as,

$$C/C_0 = (V/V_0)^{(1-\propto)} \tag{8}$$

The construction cost will not be proportional to the size of the facility due to the benefit of economies. α is considered as a scale index. The Equation (7) depicts the compost facility construction cost as being proportional to $(1-\propto)$th power of compost facility capacity. If $\propto = 0$, C is proportional to V.

Table 2. A direct comparison of total cost of waste to compost using IoT-based automation, conventional composting, sanitary landfill, and unsanitary landfill for usual 2% and 6% interest rates.

Interest Rates	Waste to Compost Cost in Conventional Facility (USD)	Waste to Compost Cost in IoT-Based Automated Facility (USD)	Sanitary Landfilling Cost (USD)	Unsanitary Landfilling Cost (USD)	Reference
2.0%	18,380,445	611,897	37,213,806	45,068,340	[57,58]
6.0%	20,385,218	1,136,695	52,206,428	60,061,558	

Table 3. A direct comparison of total cost for conventional windrow composting, decentralized IoT-based automated facilities, sanitary landfilling, and unsanitary landfilling with a 4% interest rate.

Cost Classification	Category	Conventional Composting (USD)	Decentralized IoT-Based Automated Facility (USD)	Sanitary Landfilling (USD)	Unsanitary Landfilling (USD)	References
Capital costs	Land area	14,842,820	1,542,890	554,000,000	554,000,000	[57–61]
	Automation (sensors, electronic components, software etc.)	15,350	50,000			
	Renewable power supply		25,126			
	Storage area construction (ft^2)	1000 (100 ft^2–500/ft^2)	1849.15 (100 ft^2)	12,700 (100 ft^2)	5000 (100 ft^2)	
	Urban waste collection vehicles	250,000	250,000	800,000	1,000,000	
	Electricity and water	100,000	50,000	340,460	170,230	
	Power storage		18,000			
	Inverters		14,331 (stall)			
Total (capital cost)		14,846,720	1,938,008	555,153,161	555,175,230	
Operational costs	Salary for labor per year	460,080	39,940	595,806	740,502	
	Equipment maintenance	1,953,620	11,814	383,018	500,000	
	Transportation (waste and end products)	69,168	69,168	1,702,303	3,106,909	
	Equipment depreciation	500,000	152,002	510,691	700,000	
	Utilities (Oil, lubricants, electricity, water)	28,800	3600	42,557	500,000	
Total operational cost		3,012,389	276,527	3,617,394	5,547,411	
Environmental cost	Air pollution	912	91	5,162,805	1,108,619	
Operational benefits	Value of end product	96,000	314,013	0	0	
Environmental benefits	Carbon capture credit	1668	948	1581	1581	
Total cost		3,866,001	85,711	44,315,061	52,169,878	

4.2. Scalability

The scalability of the composting operation is a crucial factor to consider when integrating IoT and related technologies into composting. Scalability refers to the ability of the system to efficiently handle large data feeds from multiple components in the composting ecosystem simultaneously, which is incredibly important when dealing with a large or very large-scale composting operation [62]. Scalability also accounts for the ability of the hardware and software of IoT, as well as related technologies integrated into composting, to handle the increased number of devices. For example, when a medium-scale composting facility transforms into a large-scale composting facility, the hardware and software in the IoT ecosystem should be able to handle the increased burden of hardware and software bulk alongside the amount of data generated without compromising the quality of the composting operation [63,64].

4.3. Technological Compatibility and Technical Expertise Availability

Assuring the technological compatibility of the chosen IoT solutions is easily compatible with the existing ecosystem of the composting facility is essential. This needs careful evaluation of the interoperability of the devices and systems in the composting facility [65,66]. Compatibility becomes essential when it directly affects compost data gathering and sharing. Harmonization among the devices used in the composting facility ensures efficiency and collaboration [67]. For example, the compatibility of the chosen sensors with the existing computer network in the composting facility would allow for a real-time data feed on the composting parameters. The availability of the technical expertise needed in installation and ongoing management is essential. For example, the composting facility must decide whether it has internal proficiency or needs external assistance [68,69]. A comprehensive understanding of the hassle-free integration of IoT and related technologies into an existing composting facility is also essential. Furthermore, any party involved in the technical aspects should be available to supervise potential issues.

4.4. Data Management Strategy: Data Privacy, Security, and Regulatory Compliance

The composting facility should draft a solid data management strategy before integrating IoT and related technologies into the composting facility [70]. This is because the vast amount of real-time data generated by the sensors and other analytics on composting should be stored, sorted, and analyzed to get used to it. For example, the facility should decide whether they need a Data Base Management System (DBMS) or a cloud computing solution. They provide scalable space for handling ever-evolving composting data [71,72]. A good database management strategy is helpful in ultimately transforming the "big data" by composing it into actionable intelligence.

The large sum of data generated due to the IoT integration into composting demands data privacy and security [73,74]. Cybersecurity measures and security audits are imperative since significant amounts of sensitive data are involved in the operations. These measures include encryption protocols, access controls, authentication mechanisms, and secure communication. For example, the confidential information of the service providers and workers and the security of the sophisticated data analytical models must be safeguarded first before the composting facility goes online. Since the composting facility should comply with the regulations implemented during the IoT integration into composting, the process of IoT integration must respect all the compliance requirements related to data management [75]. This ensures that the facility abides by ethical practices in collecting, storing, and processing large-scale datasets in composting. This ensures that the facility abides by ethical practices in collecting, storing, and processing large-scale datasets in composting. For example, since the composting facility is a business model, it is mandatory to safeguard the customers' information. Furthermore, successful adherence to regulatory requirements improves stakeholder trust [25].

4.5. Sustainability

Integrating IoT and related technologies into composting comes at a price in terms of environmental impacts [1]. One of the most essential factors the operators of the composting facility must consider is the energy consumption of the IoT systems during operation [19]. Before integrating IoT into the composting facility, the operators must consider ways to manage energy usage and mitigate the ecological footprint [76]. The sustainability of the composting operation mandates a holistic perspective of IoT deployment. This assists in reducing the environmental impact of the IoT integration into compost. Later, for integration, operators can minimize the energy usage of specific counterparts, such as connectivity devices and sensors, by analyzing their energy consumption patterns. Table 4 directly compares external pollutant emissions in IoT-based composting and conventional composting.

Table 4. A direct comparison of external pollutant emissions in IoT-based composting and conventional composting.

Category	Air Pollutant Compound	USD/Kg Emission Compound	USD/Kg Emission Compound (Present Value in 2023)[c]	Decentralized IoT-Based Automated Facility	Conventional Composting	Reference
Waste transportation	CO_2 (Fossil)	0.01	0.02	6.72	6.72	[57–61]
	CH_4 (Fossil)	0.819	0.98	0.98	0.0098	
	Particulate Matter $(PM)_{2.5}$	342	408	−0.09	0.07	
	SO_2	7.91	11.4	-	-	
	PM_{10}	2.95	3.52	−0.02	0.07	
	NO_x	4.92	7.07	-	-	
Emissions from urban waste to compost	NH_3	25.5	30.4	3.7×10^{-5}	3.7×10^{-5}	
	N_2O	11.62	13.9	0.54	0.54	
	Respiratory suspended particles	50.8	72.9	0	0	
	CH_4	0.819	0.98	−4.28	13.1	
	CO_2	0.013	0.02	−2131	1545	
	SO_2	3.97	5.71	0	0	

4.6. End Goals

A clear and concise understanding of the end goals and objectives that the operators try to achieve by integrating IoT and related technologies allows the stakeholders to understand the specific issues that can be addressed by such IoT integration in composting. This helps avoid unwanted sophistication and waste of resources where accidental introduction of IoT and related technologies might occur when fundamentally unwanted and direct the resources to segments where actual IoT integration is needed [70,77]. Defining the scope of IoT integration into composting will help product managers deliver tangible advantages over conventional methods [78]. For example, if a composting facility has an unusual number of foreign debris, such as plastic and glass, in the finished product that is not sorted in manual segregation, a computer vision-enabled waste segregation method can be used to minimize the introduction of such debris into the composting piles. After evaluating the suitability of the urban ecosystem for integrating the IoT-based composting facility, stepwise integration should be initiated. The existing literature needs more guidance for integrating the IoT-based composting facility with a modern city. The following chapter discusses the suitable steps to make the system economically and ecologically viable within an urban environment.

5. Step-Wise Integration of an IoT-Based Composting Set-Up to a Modern City and Its Ecosystem

The integration of IoT-based composting into the infrastructure of modern cities and their ecosystems takes an organizational and step-by-step approach.

5.1. Defining Requirements and the Objectives of the Composting Set-Up

Defining the explicit requirements and the scope of the operation dynamics provides a clear vantage point for defining the requirements and objectives of the IoT-based composting initiation in a city. It is essential as the generated amount of compostable waste and the type of compostable waste in modern cities, such as coffee waste, kitchen waste, yard waste, paper and cardboard, wood waste, compostable plastics, and compostable utensils, tend to be highly diverse and distinct in composition from region to region in the city. The appropriate description of the scopes of the features that will be "applied" to the composting set-up can lead the stakeholders to a more versatile, flexible, easily upgradable, and tamper-based composting set without any integration issues [79,80]. Table 5 gives an overview of the primary requirements and the objectives of the IoT-based composting system suitable for a modern urban nexus.

Table 5. Defining requirements and objectives of the intended IoT-based composting facility.

Requirement	Objective	References
Simplicity	Ability to operate and maintain the composting set-up with minimal training	[81]
Supervised decomposition	Supervised decomposition of compostable urban waste under the surveillance of IoT and related technologies	[82,83]
Energy efficiency	Minimize overall energy consumption and resource allocation	[84,85]
Pathogen detection and safety measures	Ensuring minimal health hazards and safety of the final composting product	[86]
Community engagement	Increasing the awareness and active engagement of the general public in the process of composting and its benefits on society and the environment	[87]
Enhanced durability	Recyclability and durability of the IoT components in the composting set-up for sustainability standards	[88]

5.2. Selection of Components of the IoT-Based Composting Set-Up

After a clear definition of the requirements and objectives of the composting set-up, the actual installation of the composting set-up can be initiated. It starts with properly selecting the appropriate waste bins for waste collection, segregation, and sorting, selecting sensors, data storage devices, communication protocols, etc. The selection of composting set-up components has multiple sub-steps. The process starts with the selection of the sensors to collect and transfer the data of different composting parameters, such as temperature, pH, moisture, humidity, C/N levels, and oxygen intake, which should be based on several qualities of the sensor, which makes it more applicable in IoT-based composting. They are (1) Simplicity of operation and multi-sensor capabilities: The sensor should be simple for operation and maintenance. The need for minimal training and supervision for the operators and stakeholders on the sensor activity and repair makes them economically advantageous and sustainable within an urban setup, especially in areas under rapid urbanization [81]. For comprehensive monitoring of the decomposition of the compostable waste within the facility, it is ideal if the sensors used are capable of monitoring multiple parameters and reducing the cost of operation, as well as the sophistication of the sensor integration into the composting set-up [89]. (2) Compatibility with smart systems, control, and monitoring capabilities: The sensor should be compatible with smart systems, enabling seamless integration into IoT frameworks and ensuring that data collected can be efficiently processed and utilized in a broader composting management system [84]. The sensors need to provide monitoring and control features. This refers to the capacity to control variables such as the composting process's moisture percentage, temperature, pH, and oxygen intake, which promotes effective and quick output [90]. After the sensor selection is completed, the remaining components of the composting facility should be selected step-by-step. Each sub-step of the primary component selection process is divided into consequential steps to follow in Table 6 below.

Table 6. Sub-steps of the component selection process for the IoT-based composting set-up.

Selection and Installation Step	Desirable Features/Objectives	References
Selection of other composting system components (e.g., Composting bins, compostable waste collection bins, networking cables, data storage devices, etc.).	Capable of IoT integration, made out of recyclable, reusable, and durable materials, and adhered to sustainable manufacturing process.	[82,89,90]
Choosing communication protocols	Verify interoperability with the communication infrastructure already in place in intelligent cities.	[16,84]
Implementation of sensor integration	Development of software libraries or drivers or ease of use with pre-existing ones.	[84,90–92]
Community engagement	Increasing the awareness and active engagement of the general public in the process of composting and its benefits on society and the environment	[87]
Implementation of sensor integration	Development of software libraries or drivers or ease of use with pre-existing ones.	[83,89,90]
Connectivity and networking establishment	Secure connectivity to avoid unwanted access, ability to utilize minimal power, and longer reach.	[89]
Data processing and storage	Ability to extrapolate data and give real-time analytics, faster processing, adequate storage,	[93,94]
Implement IoT security measures.	Compatibility with the latest cyber security features, accountability, and transparency of all the transactions, including monetization of compost products.	[90,93]
Cloud integration	Fast data retrievability, adequate storage ability, and seamless integration with data analysis platforms.	[90,95]
User interface and visualization	Ease of use, inter-device accessibility, and user-friendliness.	[47,90]

5.3. Integration, Testing, and Optimization with Modern City Ecosystem

The IoT-based composting ecosystem should be integrated with other operational modern city systems such as waste management, city maintenance, and infrastructure. For example, integrating the IoT-based composting system with the city's existing and operational waste management systems can ease regulatory obstacles, such as obtaining compliance certificates. The data generated by the composting facility should be integrated with the larger city platform to foster collaboration, optimize resource utilization, and enhance overall sustainability [82,96]. A robust scheme of testing and optimization should be carried out once the composting facility is online, which includes all the components, such as sensors, IoT devices, communication protocols, and control systems, by simulating diverse real-world scenarios to identify and rectify any discrepancies [90]. Configuration and optimization of all the algorithms aim to enhance composting by fine-tuning the algorithm configurations based on empirical data. Quick resolution of all the issues identified during the testing and optimization phases can ensure a reliable and stable composting process that operates seamlessly in real-world conditions. This step guarantees the composting process's effectiveness and meets the stakeholders' performance expectations [81].

5.4. Deployment, Monitoring, and Maintenance of the Composting System

During the composting facility's deployment, the components' strategic installation should be done considering the waste generation patterns and the city's population density. During the deployment, a thorough calibration of the sensors measuring composting parameters and other devices should be done to adapt to diverse environmental conditions. A robust connectivity protocol should be used to facilitate the real-time monitoring of the decomposition while adhering to stringent security measures to protect against cyber threats. A comprehensive testing and validation scheme should safeguard the composting process's reliability [82]. Continuous monitoring and proactive maintenance of the composting sensors and other components of the composting facility are vital for real-time data analytics of the decomposition of compostable waste. It is vital to have an alert system to identify and notify the relevant parties of critical deviations and facilitate swift responses to potential issues. For example, a malfunction of the sensor readings, etc. The plan for monitoring and maintenance should encompass routine checks, inspections, and predictive strategies [82,97]. In order to reduce the environmental impact, ensure the long-term viability during installation and integration of the IoT-based composting facility, and adhere to sustainability standards. The following chapter discusses sustainability standards when installing and integrating the IoT-based composting system in a modern urban city. Figure 6 gives an overview of the strategic flow path step-by-step installation of an IoT-based composting facility in a modern city ecosystem.

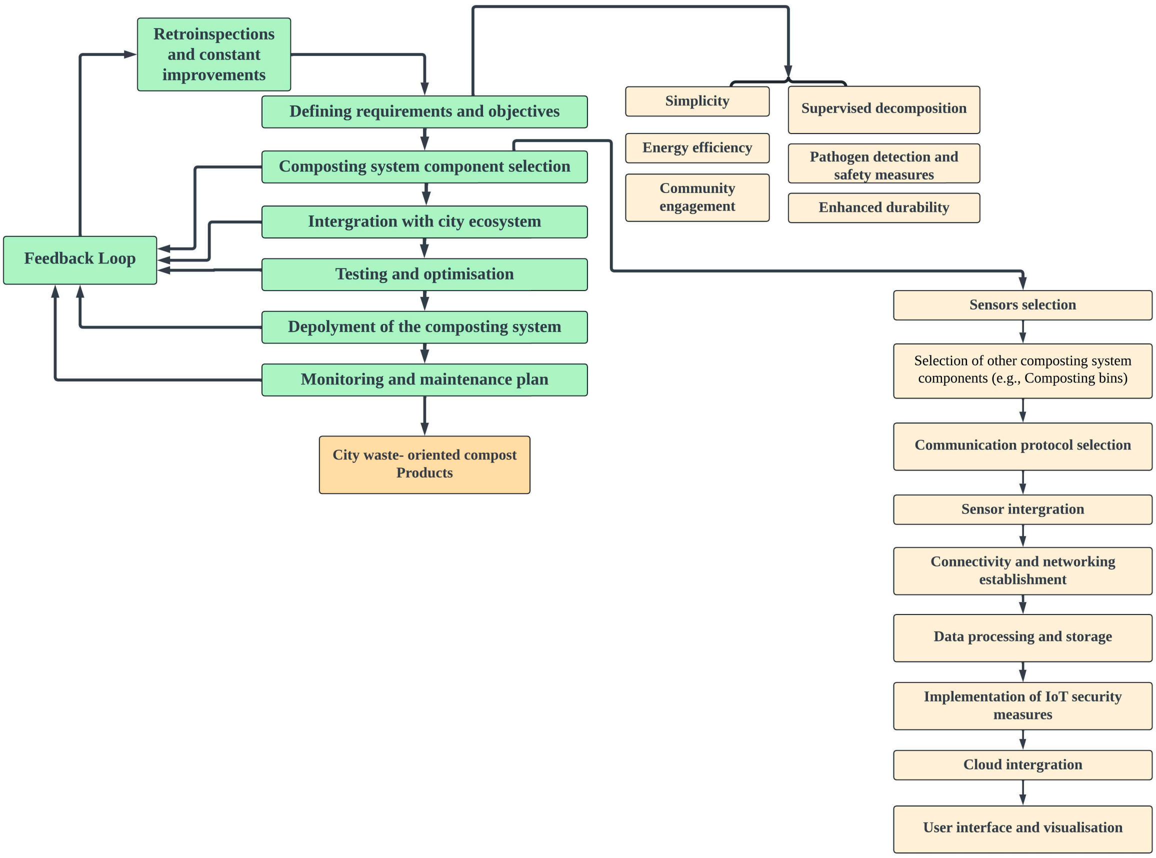

Figure 6. A strategic flow path step-by-step installation of an IoT-based composting facility in a modern city ecosystem.

6. Guidance for Achieving Sustainability on Sustainable Development Goals (SDGs) During Deployment of the IoT-Based Composting Facility

The overall strategic plan proposed in this article is based on SDG 11, Sustainable Cities and Communities, and SDG 13, Climate Action. The final goal of this strategic plan is to successfully establish a sustainable community within modern city limits with city-wide compostable waste sustainability. The city-wide waste sustainability and the enhanced recyclability, reusability, and responsible management of assets in this strategic plan ultimately support the climate action to reduce global pollution. Nevertheless, the article provides detailed actions below to achieve most SDGs via compartmentalization of the strategic plan by adhering to a sustainability-oriented approach.

6.1. Life Cycle Assessment (LCA): SDG 12—Responsible Consumption and Production

Performing an LCA for the IoT-based sensors, other electronic equipment such as data storage disks, and mechanical devices such as impellers used to mix compost is vital to attaining sustainability within the composting ecosystem. According to [98], performing an LCA on IoT-based sensors and other mechatronic devices of the composting facility can be used to make a life cycle modelling framework that can be later used for recovering the usable materials, such as carbon and metals, for the manufacture of new devices for the composting facility. The emission of gases such as ammonia and methane during compost production is often overlooked. Therefore, assessing the byproducts of compost production during the production phase of the life cycle, such as the gaseous emissions, should be retained or minimized to reduce the environmental impact (greenhouse effects) [94,99]. Furthermore, LCA can assist in determining whether the final product of compost contains any toxic substances, such as heavy metals, and take action to remedy those toxic products via suitable sustainable solutions, such as an early contaminant detection system to identify any foreign matter, such as small-scale electronic waste or plastic products that can be mixed with compostable waste [100,101]. LCA can give unbiased feedback on the entire composting life cycle and pinpoint loopholes that might threaten the environment and the community. For example, LCA can assist the stakeholders in improving and restructuring the bioreactors of the composting facilities to capture gaseous emissions and attain zero emissions standards. LCA may be used to optimize the design and manufacturing processes of IoT devices used in composting systems for IoT-based composting. This entails choosing components with less of an adverse effect on the environment and building long-lasting and recyclable equipment [102,103]. LCA can give an insight into the components with a high environmental footprint in the composting ecosystem and replace them with low-embodied energy materials (e.g., replacing plastic compostable waste collection bins with stainless steel and more durable bins). LCA is useful for determining where resource use might be minimized. This might entail maximizing the composting set-up's total resource efficiency, material selection, and energy utilization in the Internet of Things-based composting context. Furthermore, it assists in the selection and comparison of composting technologies based on their environmental friendliness. Additionally, it directs the development of Internet of Things-based composting systems by spotting chances for innovation and technical breakthroughs. In summary, LCA for IoT-based composting ensures a systematic and holistic approach to sustainability, guiding the development, implementation, and ongoing improvement of composting technologies in urban settings.

6.2. Energy Efficiency: SDG 7—Affordable and Clean Energy

The use of forced aeration in composting piles of the facility is vital for the proper degradation of the waste and transformation into compost. As this process is automated and mechanically powered, it requires energy that can be sustainably obtained to reduce operational costs. For example, solar energy is used to power the composting facility's sensors and equipment and reduce the operation's gas emissions by replacing the gas generators used for mechanically aerating the compost piles [69]. Alternative energies such as thermal collectors, wind turbines, and biogas energy, other than fossil fuels, can be effectively used to power up the electronic components of the composting facility, improving energy efficiency and sustainability [104]. Furthermore, the design of the composting sensors and data collection devices for maximum energy efficiency can minimize the power consumption of these devices during their active usage within the composting facility. Furthermore, it is essential to implement sleep or low-power consumption modes when these devices are not collecting data. For example, intelligent composting bins for collecting compostable waste can be set up on sleep or low power modes when indicated empty [105].

6.3. Materials Recyclability Andreusability: SDG 12—Responsible Consumption and Production and SDG 9—Industry, Innovation, and Infrastructure

According to [26], the precious metallic components in IoT-based sensors and other electronic equipment can be extracted and reused in IoT-based sensors and other devices in the composting facility. For example, end-of-life electrical conductivity sensors and printers can extract carbon for remanufacturing electrical conductivity sensors for the composting facility [106]. The design of IoT devices, such as sensors, data storage units, and compostable waste collection bins for easy disassembly, saves labor, energy, and resources for recycling such end-of-life components in the composting facility [107]. Using reusable components in the composting ecosystem reduces the need for frequent replacements. For example, long-lasting and reusable polymers in compostable waste collection bins can manufacture circuit boxes for IoT devices and insulation shields for composting vessels within the composing facility [106].

6.4. Local Sourcing and Manufacturing: SDG 8—Decent Work and Economic Growth

According to [108], sourcing localized services and using local resources for the design, maintenance, and management of the IoT-based composting facility improves the awareness and operational control of the composting facility. It optimizes the limited natural resources within the city limits. Another advantage of the usage of localized services and resources in the IoT-based composting facility is that the authorities get the possibility to identify and disseminate information properly prior to any crisis related to compostable waste management before any crisis starts and intensifies (e.g., complaints on improper and irregular compostable waste collections). Local sourcing of the materials and services for the composting facility can reduce congestion, provide employment opportunities to the local community, reduce transportation costs of compostable waste from collection centers to the facility, and reduce the environmental impact [109]. Furthermore, manufacturers (e.g., composting parameter sensor manufacturers) and suppliers (waste collection companies) should adhere to sustainable practices such as certifications for environmental and social responsibility [110].

6.5. End-of-Life Considerations: SDG 12—Responsible Consumption and Production

A solid plan to dispose of end-of-life IoT devices from the composting operation at the end of their life cycle is essential to meet sustainable standards. The end-of-life IoT devices, or e-waste, of composting operations are critical challenges that must be addressed. The development of IoT devices, with enhanced life spans and the development of disposal strategies for old devices such as composting sensors, are essential to enhance the sustainability aspects of the IoT-based composting operations, along with the legal framework to safeguard proper disposal of IoT devices (e.g., sensors, data storage drives, etc.) in the composting operation. For example, the European Union Waste Framework Directive is a legal framework that promotes responsible disposal of IoT devices that can be easily integrated into IoT-based composting facilities [111]. According to [112], it is essential to deploy evolving prediction algorithms to determine the needs and expenses incurred in the future regarding the infrastructure of the IoT-based composting system to ensure a proper upgrade of the composting operation without provoking any environmental damage or unwanted discharge of electronic and other wastes due to unplanned, unorganized, and chaotic repairs and upgrades. The resources for producing new IoT devices for composting facilities can be obtained by mining the e-waste repositories in an urban setting and promoting resource recycling and circulation. For example, end-of-life sensors can be used to recover carbon and iron to manufacture new IoT sensors and other electronic devices for composting facilities. This facilitates a closed-loop circular economy within the IoT-based composting ecosystem [113].

6.6. Community Goals and Education; SDG 4—Quality Education

The community directly interacting with the IoT-based composting facility should be educated about the advantages of responsible waste management and composting and their impact on the environment and science. Delivery of these educational insights to improve community engagement in composting is successful when delivered at every level of education (practical sessions at the IoT-based composting facility and theoretical ones such as seminars and webinars on the importance of composting and city-wide waste sustainability) [109]. One of the best ways to educate and get active community engagement to achieve the sustainability of composting operations is by conveying and communicating with the local communities on the advantages of composting in an urban setting. Making the community a part of the body that makes decisions in terms of city-wide waste management, co-creation, and deployment of different technological tools for composting, such as a user experience survey of a new compostable waste collection mobile application and data exchange, can establish a good quality of life in the community where the waste is systematically removed, recycled, and given back to them as a value-added product (compost) [114].

6.7. Inclusivity and Accessibility: SDG 10—Reduced Inequalities

Inhabitant sustainability of the IoT-based composting operation of modern cities is paramount to securing inclusivity and the accessibility of the composting service to all the hierarchies of urban society. This can be achieved by decreasing the digital divide and equal opportunities by allowing low-income communities to enjoy compostable waste management services as a collective hub (e.g., a common waste collection center, a collectively accessible compostable waste collection routine updates). It is essential to ensure that compostable waste management facilities are accessible to all the community members, including those who are differently abled. With the Sustainable Development Goals (SDGs), IoT-based composting operations in modern cities are mandatory to evolve with inclusive solutions with enhanced accessibility to all sectors of the urban society. For example, the development of UIs for the IoT-based composting system should consider the diverse needs and preferences of all the urban community members [109].

6.8. Biodiversity and Ecosystem Health: SDG 15—Life on Land

According to [115], integrating IR 4.0 and 5G technologies with city-wide waste management and IoT-based composting operations leads to many unresolved natural calamities. For example, developing IoT-based composting sensors and database management systems with a lowered carbon footprint is prolonged [116]. Proper legislation and standard mechanisms are essential for the disposal of the End-of-Life IoT-based sensors and other electronic counterparts discharged from the IoT-based composting set-up. According to [117], biodiversity preservation in IoT-based composting ecosystems can be done by integrating the knowledge of urban ecology and the effects of urbanization on biodiversity during the planning, development, and maintenance of the IoT-based composting set-ups. Integrating waste segregation settings during compostable waste collection to detect plastic, glass, and hazardous waste can minimize the environmental impact of IoT-based composting operations.

6.9. Continuous Improvement: SDG 9—Industry, Innovation, and Infrastructure

According to [118], the active participation of the citizens in the modern city in the IoT-based composting setup can lead to collaborative experiences and improve service in compostable waste management. Establishing a feedback loop mechanism can improve IoT-based composting depending on the feedback of stakeholders, operators, government authorities, and the general public. According to [119], 360-degree feedback can be deployed to measure and evaluate the level of satisfaction of different stakeholders (citizens, workers, government authorities, etc.). The anonymity of the 360-degree feedback mechanism will remove any drawbacks and loopholes in the current IoT-based composting ecosystem.

By integrating these sustainability principles into the development and implementation of the IoT-based composting system, stakeholders can contribute to the overall sustainability goals of the modern city ecosystem. Regular monitoring, evaluation, and adaptation based on evolving sustainability guidelines will be crucial for long-term success. Figure 5 gives an overview of the sustainability criteria that should be considered during IoT integration into a modern city ecosystem. Figure 7 gives an overview of the achievement of Sustainable Development Goals (SDGs) via the proposed strategic plan in city-wide IoT-based composting in an urban ecosystem. Table 7 gives an overview of the contributions to SDGs by IoT technologies in composting processes based on the essential counterparts of the proposed IoT-based composting system introduced in this article.

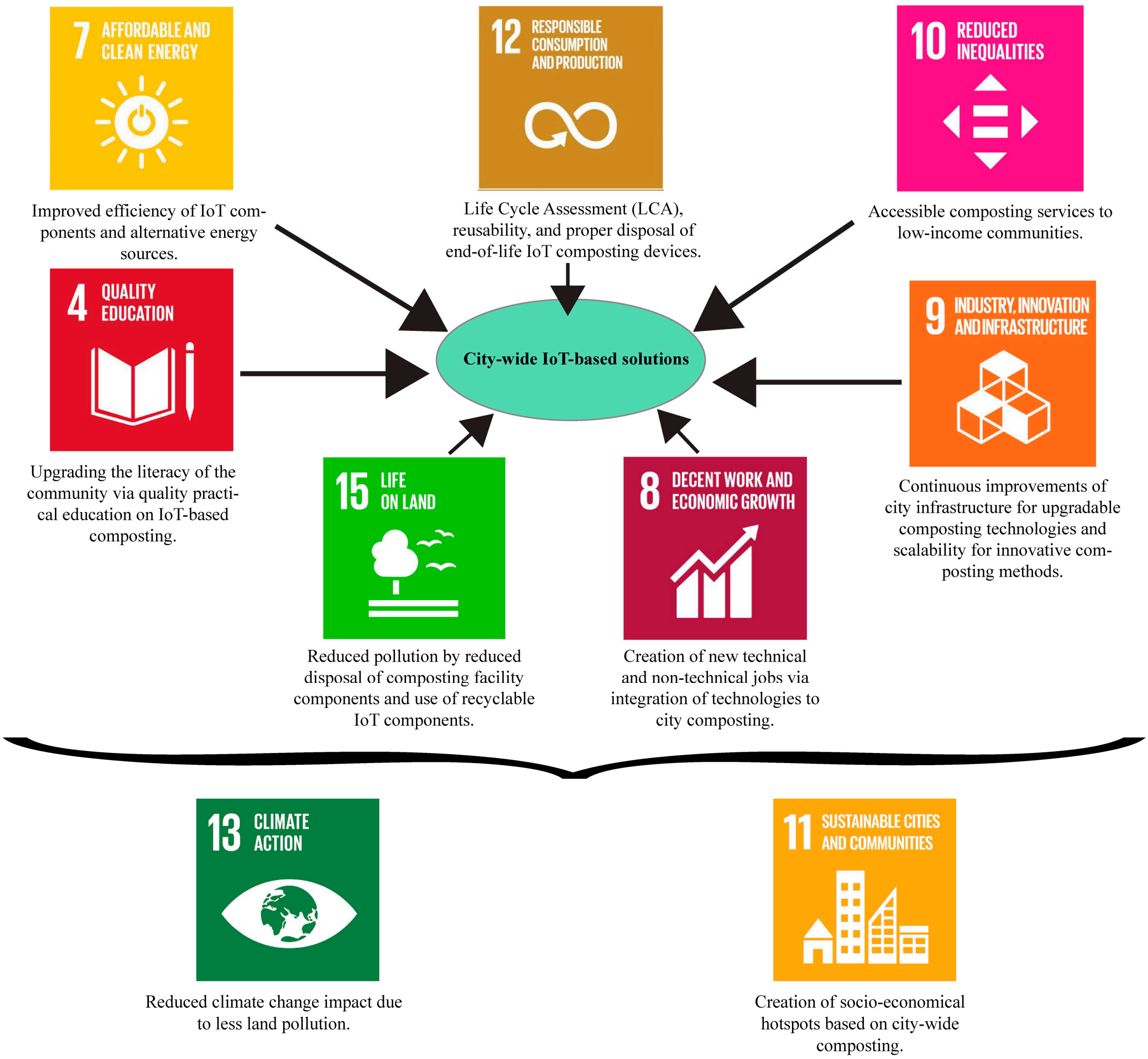

Figure 7. Achievement of Sustainable Development Goals (SDGs) via the proposed strategic plan in city-wide IoT-based composting in an urban ecosystem.

Table 7. Contributions to SDGs related to the contributions of IoT technologies in composting processes.

Proposed Composting Systems Component	Contribution to SDGs	Examples of Real-World Contribution	Reference
Composting Parameter Sensors	SDG 12 (Responsible Consumption and Production)	Uses machine learning-based moisture sensors to monitor the moisture content of composting in real time and uses self-adjustment to control the moisture content to improve product quality	[28]
Connectivity	SDG 9 (Industry, Innovation, and Infrastructure)	Utilizes highly robust Wi-Fi connectivity to monitor the composting process and real time waste segregation using convolutional neural networks to accelerate production	[23]
Data Collection and Analytics System	SDG 12 (Responsible Consumption and Production)	Utilization of near infrared spectroscopy (NIR) to accurately predict moisture content, total carbon, carbon/nitrogen ration and organic matter content during composting and accurately predict compost maturity and save time	[120]
UI	SDG 12 (Responsible Consumption and Production)	Uses 3D object detection model based on convolutional neural networks to detect and classify compostable and non-compostable waste in real time to reduce contamination and product pollution	[121]
Automation and Control System	SDG 13 (Climate Action)	Utilization of node MCU (micro-controller unit) to automate amount of feedstock, composting process, moisture content and reduce odor emissions to reduce air pollution	[73]
Security System	SDG 12 (Responsible Consumption and Production)	Utilization of middleware and cloud and edge-abstraction layers for secure communication of composting data to reduce data theft	[82]
Power Supply and Backup	SDG 7 (Affordable and Clean Energy)	Conversion of organic waste into compost within 4 weeks of time and maintaining optimal 40% moisture and 65% operational efficiency in solar power via photovoltaic energy storage system to reduce the dependence on fossil fuel energy	[21]

7. Discussion

The primary strategic plan presented in this article can vary depending on several factors, such as the geopolitical landscape of the city, the Gross Domestic Production (GDP), and individual income levels of the modern city. Cities with higher GDP can adapt to a highly technical and sophisticated IoT-based composting platform with fewer issues in scalability and budgetary constraints. Nevertheless, cities with lower GDPs must adhere to a less sophisticated IoT platform, vital for creating employment opportunities. Therefore, the thorough examination of the economic markers of a city prior to the execution of the plan is vital to its success.

8. Future Directions

The successful integration of IoT into modern city composting can lead to "waste conversion hotspots". The novel concept introduced by authors here can simulate a fully functional eco-industrial complex where a majority of the economy of a particular urban setting is dependent on dedicated IoT-based composting facilities where families reside and engage in economic activities as employees as well as users of the IoT-based composting facility. Therefore, future strategic plans for IoT-based composting in modern cities are recommended to include stimulants for the rapid proliferation of waste conversion hotspots near the IoT-based composting facility.

9. Conclusions

A fragmented research landscape in IoT-based composting in modern city ecosystems is revealed in the systematic review of publications. There is a critical need for a scalable and integrated framework for the full implementation and execution of urban IoT-based composting in modern cities. Even though IoT is used in real-time monitoring, automation, and intelligent composting optimization, most of the studies focus narrowly on isolated aspects with limited practical urban applications. IoT-based composting provides clear advantages over conventional urban composting in areas such as enhanced monitoring capabilities, efficiency, resource utilization, and management. There is insufficient strategic guidance for the comprehensive urban scalability of IoT-based composting. The review provides cohesive strategic guidance to incorporate cost-effective composting components, real-time composting data management, and compliance considerations for sustainable development.

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